Edexcel A Level Business

Your notes

2.4 Resource Management

Contents

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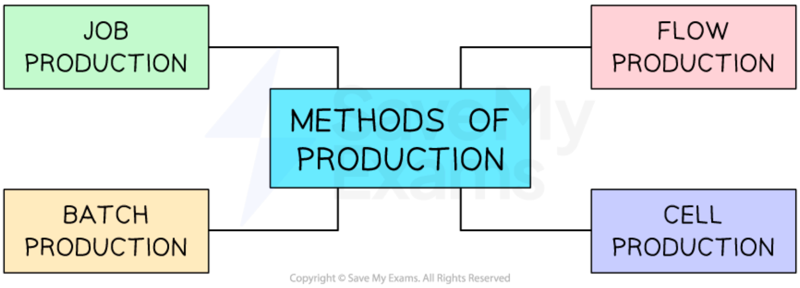
Methods of production

- Production is the **transformation of resources** (e.g. raw materials, components and processes) into finished goods or services
    - Goods are **physical products**, such as bicycles and T-shirts
    - Services are **non-physical items**, such as hairdressing, tourism and manicures
- Businesses can **organise their production** processes in a variety of ways

The main methods of production



The main methods of production are job, batch, flow and cell production

- The method of production used by a business will depend on a number of factors, including:
    - The **level of output** required to be produced
    - The **nature** of the product
    - Whether the product is **standardised or customised**
    - The level of **automation** used in production

Job production

- Producing **one item at a time**, as ordered by the customer
    - Advantages
        - **High-quality** product
        - Motivated and **highly skilled workers**
        - Customised products can be produced
    - Disadvantages
        - Production is **slow**
        - **Labour costs** are high

Batch production

- Groups of the same product are produced before moving on to a group of different products
  - Advantages
    - Workers can **specialise**
    - Production can take place as the previous batch starts running out
  - Disadvantages
    - **Requires careful coordination** to avoid shortages
    - Money is **tied up in stock**, as completed products need to be stored

Flow production

- **Continuous manufacturing** of standardised products, usually on a production line
  - Advantages
    - Low unit costs due to **economies of scale**
    - Rapid production
    - Usually highly automated (**capital-intensive**)
  - Disadvantages
    - Customisation is **difficult**
    - Capital equipment can be **expensive to purchase**

Cell production

- This involves workers being organised into **multi-skilled teams**, with each team responsible for a **particular part of the production process**
  - Advantages
    - Cell production is often more efficient than other methods, as workers share their skills and expertise
    - Motivation is usually high, as employees work as a team
  - Disadvantages
    - Requires extensive reorganisation of production processes
    - Teams' efficiency may be reduced by weaker workers

Worked Example

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Blush Cosmetics uses batch production to make its range of soaps and bath bombs, with a variety of unique ingredients and scents. The business sells its products in its own high street store and online and has recently started supplying its products in small quantities to a chain of exclusive hotels.

Explain one likely reason why Blush Cosmetics chooses to use batch production. [4]

Step 1: Identify a reason for the business to use batch production

One reason for Blush Cosmetics to use batch production is that it allows groups of products with varied ingredients to be manufactured. **[K]**

Step 2: Include a reference to the business scenario

Blush Cosmetics sells a range of bath bombs and soaps that have different fragrances, colours and ingredients. **[Ap]**

Step 3: Develop the reason using a connective

Blush Cosmetics needs to produce significant levels of output to meet increasing demand and batch production, as well as to provide the opportunity to change ingredients between batches **[An]**, which can allow quite large quantities to be produced for the company's high street and online stores **[An]**.

Or:

Using batch production can allow Blush Cosmetics to produce smaller quantities of unique products for exclusive hotel customers **[Ap]**, meeting their specific needs **[An]**.

Examiner Tips and Tricks

Carefully consider the needs of the customers to whom a business sells when recommending a suitable method of production.

Where the selling price is a key driver of consumer demand, flow production (where **unit costs** are minimised) is likely to be very suitable.

Where demand is driven by quality or where customisation is required, job production or batch production is likely to be a better choice.

Calculating productivity

- **Productivity** is the output per input (person or machine) per hour (e.g. an IKEA worker is able to produce two Poäng chairs per hour)
  - This is not production, which is the **total amount of output produced** in a time period (e.g. IKEA produced 300 Poäng chairs in February)

Labour productivity

- The **labour productivity** of a business is a measure of the **output per worker** during a specified period of time
- Labour productivity is calculated using the formula

$$\text{Labour productivity} = \frac{\text{Output}}{\text{Number of workers}}$$

Capital productivity

- **Capital productivity** is a measure of the **output of capital employed (e.g. machinery)** during a specified period of time
- Capital productivity is calculated using the formula

$$\text{Capital productivity} = \frac{\text{Output}}{\text{Number of machines}}$$

Worked Example

The table shows the number of pairs of luxury wool socks produced by Scotty Socks Ltd in 2021 and 2022.

Year	Units produced
2021	46,000
2022	69,000

In 2021, Scotty Socks employed 50 staff. In 2022, the number of staff employed by the business increased by 20%.

Calculate the percentage change in labour productivity between 2021 and 2022.

[4]

Step 1: Calculate the labour productivity for 2021

$$= \frac{46{,}000 \text{ units}}{50 \text{ workers}} \quad \textbf{[1]}$$

$$= 920 \text{ units per worker}$$

Step 2: Calculate the labour productivity for 2022

$$= \frac{69{,}000 \text{ units}}{60 \text{ workers}} \quad \textbf{[1]}$$

$$= 1{,}150 \text{ units per worker}$$

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Step 3: Calculate the percentage difference between the two years ([new–old] ÷ old)

$$= \frac{1,150 - 920 \text{ units}}{920 \text{ units}} \times 100 \quad \textbf{[1]}$$

$$= 25\%$$

Step 4: Identify whether the percentage difference is an increase or a decrease

25% **increase [1]**

Worked Example

Rolvo Ltd forecasts that by the end of the year, it will produce 250,800 units and that capital productivity will be 1,100 units per machine.

Calculate the number of machines Rolvo Ltd has in use.

Step 1: Divide the output by the capital productivity

$$= \frac{250,800 \text{ units}}{1,100 \text{ units}}$$

$$= 228 \text{ machines}$$

Factors that influence productivity

- The productivity within a business can often be improved
- When productivity increases, **business costs decrease**
- When business costs decrease, the firm can either pass on this decrease to consumers in the form of **lower prices** or maintain the selling prices and enjoy higher **profit margins**

Factors that influence productivity

Factor	Explanation
Employee motivation	▪ Motivated workers tend to be more productive ▪ **Financial incentives linked to output** may increase worker productivity ▪ A non-financial incentive may be **including workers in decision-making**, which will increase their commitment and productivity

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Skills, education and training staff	▪ Well-trained and educated workers are likely to be **able to make useful contributions** to decisions that improve productivity
	  ▪ Workers are more **autonomous**, so the need for supervision is reduced
	  ▪ Contributions from knowledgeable staff are likely to lead to **improvements in productivity**

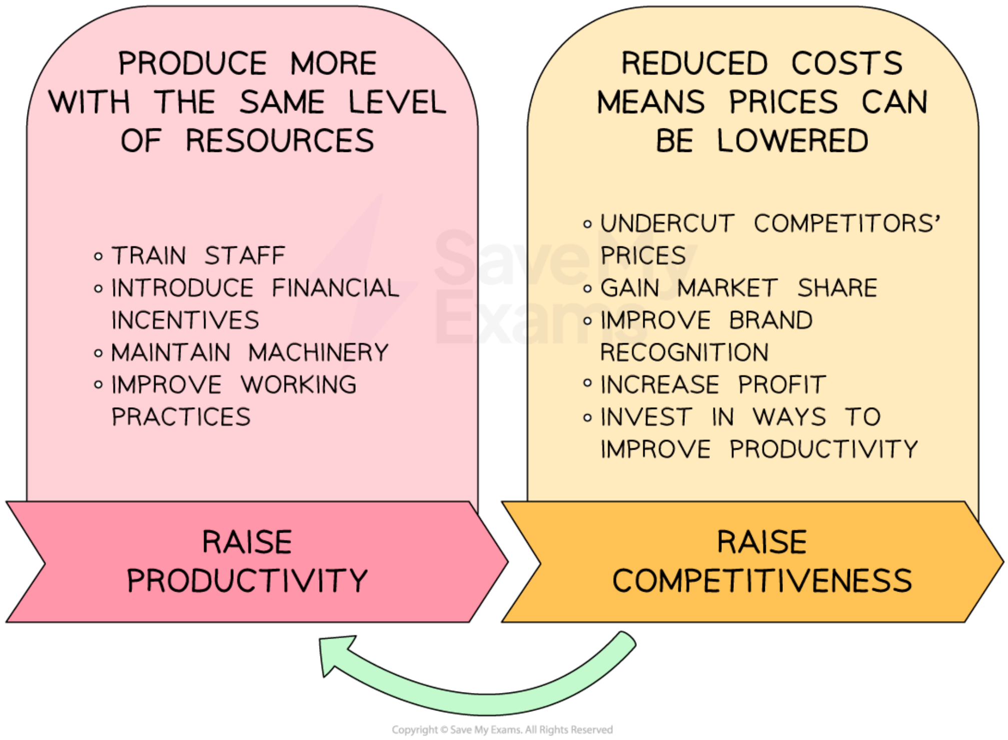
Business organisation and working practices	▪ Flexible and adaptable workplaces can **improve the commitment of workers** and allow a business to **respond to changes in demand** ▪ **Hours and locations of work** can be adapted to better meet the needs of workers and the demand ▪ **Workstations may be used for various purposes** with careful planning and training

Investment in capital equipment	▪ Increased automation can **improve levels of output and quality**
	  ▪ Well-chosen machinery is **less likely to make mistakes** than humans
	  ▪ Machinery and technology can **operate for long periods without a break**, as long as they are properly maintained

The link between productivity and competitiveness

- Competitiveness refers to the ability of a business to maintain or grow its sales and market share, **given the presence and actions of rivals**
- Businesses that increase their level of productivity (e.g. of workers or capital equipment) are likely to be **more competitive**

Productivity and competitiveness



There is a strong link between productivity and business competitiveness

- Businesses that are competitive are likely to have the **financial resources** required to continue investing in improvements to their productivity

Understanding efficiency

- Efficiency refers to the ability of a business to **use its production resources** as cost-effectively as possible
  - Efficiency is often measured in terms of the **average cost per unit**
  - The average cost per unit is calculated using the formula

$$\text{Average cost per unit} = \frac{\text{Total costs}}{\text{Number of units}}$$

- **Maximum efficiency** is achieved when the cost per unit is at its lowest

Diagram showing maximum efficiency

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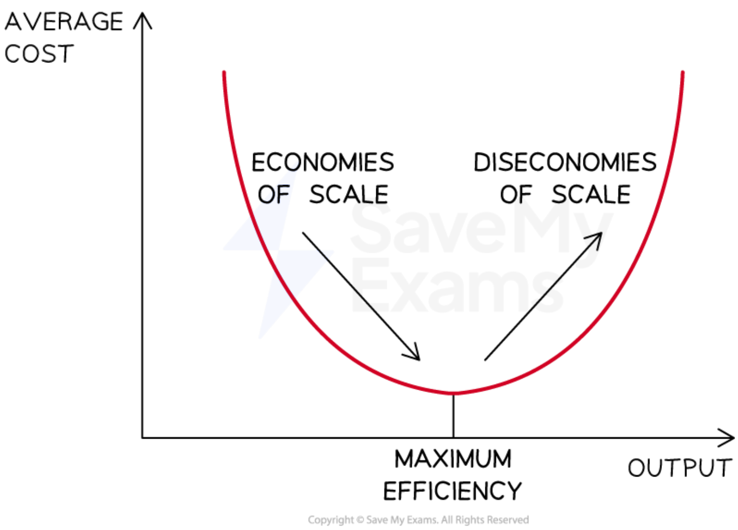

Graph showing that maximum efficiency occurs at the point where the costs are at their lowest

- The average cost curve shows that the most efficient level of production is achieved when:
  - **Economies of scale** are maximised
  - Total costs are spread across an optimum level of output
  - **Diseconomies of scale** are minimised

Factors that influence efficiency

Factors that can influence business efficiency

Factor	Explanation
Standardisation of the production process	▪ Occurs when all staff use the **same components and techniques** in the production process ▪ **Training of workers** is minimised ▪ **Bulk buying** of components reduces variable costs ▪ Production **lead time is reduced** ▪ **But** customisation of products is not usually possible
Relocation or downsizing	▪ Moving production to a **cheaper or smaller location** can reduce **fixed costs** ▪ **Labour-intensive** businesses may look for **lower-wage locations** ▪ Having capital-intensive locations may mean looking for **lower rents or land costs**

	▪ However, relocation is very disruptive and will incur significant short-term costs
Investment in capital equipment	▪ **Purchasing or upgrading machinery and technology** can increase the rate of output, lower costs and improve quality
Organisational restructuring	▪ **Reducing the level of staff** or **reorganising** staff can better match labour to output needs ▪ **Delayering reduces labour costs**, as levels of management are removed ▪ Redeployment can **motivate workers** by providing opportunities for staff to take on a new role, which will develop their skills and experience
Outsourcing	▪ Tasks may be given to **other specialist businesses** that can complete them at a lower cost ▪ Outsourcing allows a business to focus on improving the efficiency of its **core competencies**
Adoption of lean production techniques	▪ An approach to production that involves the **reduction of all types of wastage** (time, resources and space) ▪ **Kaizen** means that **improvements are made continuously** ▪ **Just-in-time** involves **holding little or no stock**, which minimises storage costs

Capital-intensive and labour-intensive production

- **Labour-intensive production** predominantly **uses physical labour** in the production of goods/services
  - The **delivery of services** is usually more labour-intensive than manufacturing
  - In countries where **labour costs are low** (e.g. Bangladesh), labour-intensive production is common
  - **Small-scale production** is likely to be labour-intensive
  - For example, UK schools are labour-intensive operations, as teachers plan and deliver lessons and provide pastoral support
- **Capital-intensive production** predominantly **uses machinery and technology** in the production of goods and services

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- **Large-scale** production of **standardised products** is likely to be capital-intensive
- Manufacturing in **developed countries**, where **labour costs are relatively high**, is likely to be capital-intensive
- For example, automotive manufacturers such as Ford use robots and other production technology to manufacture vehicles, with supervisors overseeing the quality of output

Your notes

Evaluation of labour-intensive and capital-intensive production

Type of production	Advantages	Disadvantages
Capital-intensive	▪ **Low-cost production** where output is high ▪ Machines are usually consistent and precise ▪ Machines can **run without breaks**	▪ Significant set-up and **maintenance costs** ▪ **Breakdowns** can severely delay production ▪ May not provide **flexibility** in production
Labour-intensive	▪ **Low-cost production** where labour costs are low ▪ Provides opportunities for workers to be creative ▪ Workers are flexible (e.g. they can be retrained)	▪ Workers **may be unreliable** and need regular breaks ▪ **Incentives** may be needed to motivate staff ▪ **Training costs** can be significant

Calculating capacity utilisation

- Capacity utilisation is a **measure of the level** to which a business's **assets** are being used to produce output
- It **compares the current output to the maximum possible output** a business can produce using all of its assets and is expressed as a percentage
- Capacity utilisation is calculated using the formula

$$\text{Capacity utilisation} = \frac{\text{Current output}}{\text{Maximum possible output}} \times 100$$

Worked Example

Lola Bakery produces specialist Indian and Bangladeshi breads that are sold to restaurants in the West Midlands. Batch production is used in the factory to manufacture a range of breads, and the factory can produce a maximum of 68,400 units per month.

In May, factory output was 51,420 units.

Calculate Lola Bakery's capacity utilisation in May.

Step 1: Divide the current output by the maximum output

$$= \frac{51{,}420 \text{ units}}{68{,}400 \text{ units}}$$

$$= 0.75$$

Step 2: Multiply the outcome by 100 to obtain the percentage capacity utilisation

$$= 0.75 \times 100$$

$$= 75\%$$

Worked Example

Lola Bakery produces specialist Indian and Bangladeshi breads that are sold to restaurants in the West Midlands. Batch production is used in the factory to manufacture a range of breads, and the factory can produce a maximum of 68,400 units per month.

In August, capacity utilisation increased to 92%.

Calculate the number of units produced by Lola Bakery in August.

Step 1: Convert 92% to a decimal

$$= 92\% \div 100$$

$$= 0.92$$

Step 2: Find 92% of the maximum output using the decimal

$$= 0.92 \times 68{,}400$$

$$= 62{,}928 \text{ units}$$

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Implications of underutilisation and overutilisation of capacity

Underutilisation of capacity

- If a business has a low level of capacity utilisation, it will **not be making the most of its resources** and is likely to have **increased unit costs**
  - The **fixed costs** are spread over fewer units of output, resulting in higher average total costs
  - Workers may be **underdeployed**, leading to fears of **redundancy**
- Operating under capacity does provide a business with **flexibility**
  - There may be an opportunity to engage workers in **maintenance tasks**
  - The business can respond to **sudden increases in demand**

Overutilisation of capacity

- If a business has a high level of capacity utilisation, it may not have the **flexibility to respond to new orders** from customers
  - Staff will be under a lot of pressure to produce high levels of output
  - Overworked staff may be inclined to leave, increasing **staff turnover**
  - Machinery may be pushed to its limits and prone to **breakdowns**, which **disrupts production and increases costs**
- High capacity utilisation will **minimise average total costs** and increase business **competitiveness**
  - If workers are busy, they are likely to **feel secure in their employment**
  - A business that is busy is likely to be **well thought of** by customers and will attract customers who are willing to wait for products to be delivered

Worked Example

Production data for pencil manufacturers A and B

Manufacturer	Current output (units)	Maximum output (units)
A	3m	6m
B	2.4m	4m

Explain **one** implication of the level of capacity utilisation for pencil manufacturer A, compared to manufacturer B. You are advised to show your workings.

[4]

Step 1: Demonstrate knowledge of capacity utilisation

- Capacity utilisation measures how well a business is using its available production capacity. **[1]**

Step 2: Calculate the capacity utilisation for both manufacturers

$$\text{Manufacturer A} = \frac{3 \text{ million units}}{6 \text{ million units}} \times 100$$ **[1]**

$$= 50\%$$

$$\text{Manufacturer B} = \frac{2.4 \text{ million units}}{4 \text{ million units}} \times 100$$ **[1]**

$$= 60\%$$

Step 3: Explain the implications

One implication is that staff may be bored and demotivated, particularly at manufacturer A, because they have less work to do than at manufacturer B. **[An]**

Or:

Unit costs are likely to be higher than they could be at both manufacturers **[K]** because resources such as workers and machinery are not being used to their full potential **[An]**.

Examiner Tips and Tricks

Including a definition of the key term is not essential, as **implicit understanding** of a business term or concept can be shown through correct calculations and identifying appropriate points.

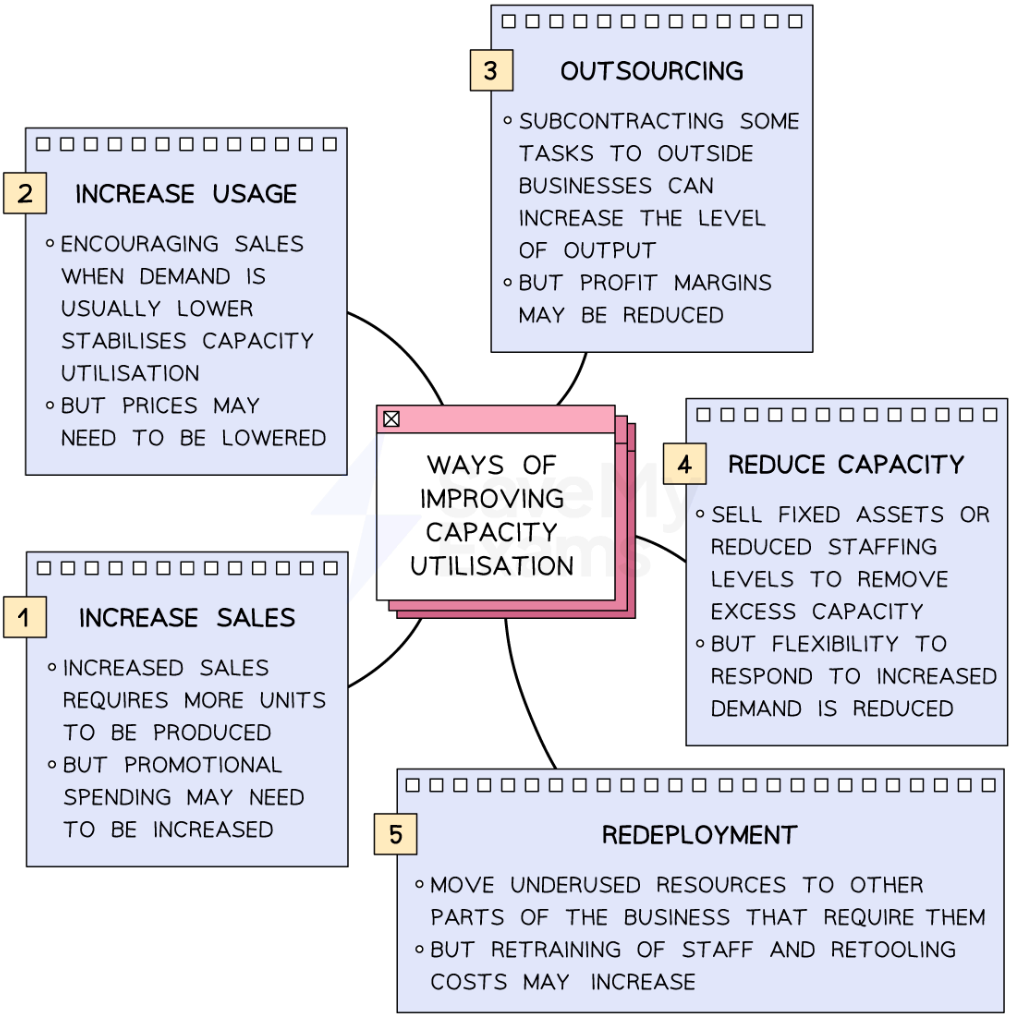
In the example above, the definition could attract a mark if calculations had not been included, but the answer achieved full marks because it included **correct calculations** as well as an **appropriate point** with an **explanation**.

Your notes

Ways of improving capacity utilisation

- Achieving an **optimum** level of capacity utilisation is a key **production objective**
- A business has several options available to improve its **capacity utilisation**

A range of ways to improve capacity utilisation



Ways of improving capacity utilisation include increasing sales, reducing capacity and using outsourcing or redeployment

- Increasing the level of capacity utilisation results in lower unit costs
- In some cases, **increasing demand and thus output** is the most suitable way to increase capacity utilisation
  - For example, reducing selling prices or offering short-term **price promotions** can increase demand

- If **competitors go out of business**, demand should pick up, which will improve capacity utilisation without the need for further promotion
- **Reducing the level of overall capacity** will improve the level of capacity utilisation
  - Capacity can be reduced by selling capital equipment or reducing the number of staff

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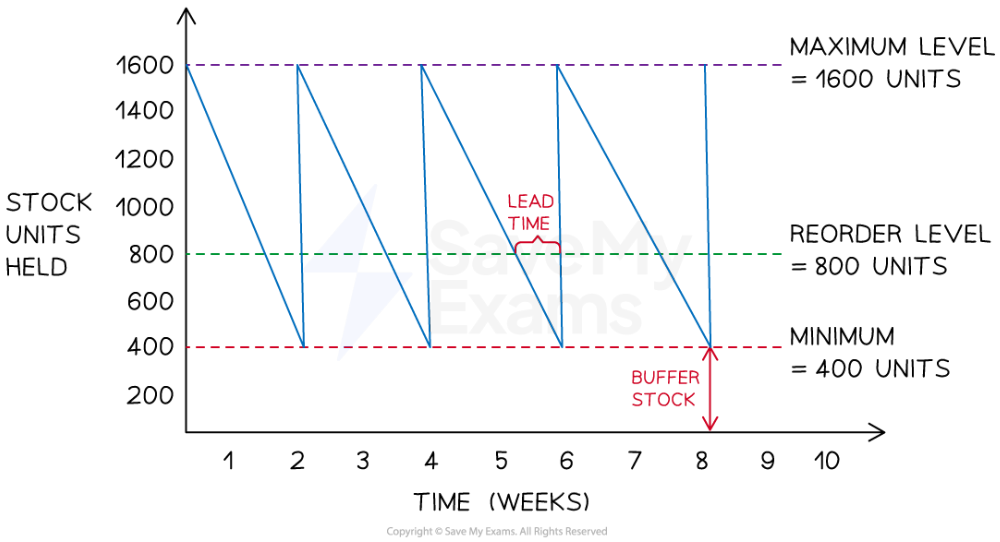
Interpretation of stock control diagrams

- A stock control diagram illustrates the **flow of stock** (inventory) into and out of a business over time

Example stock control diagram



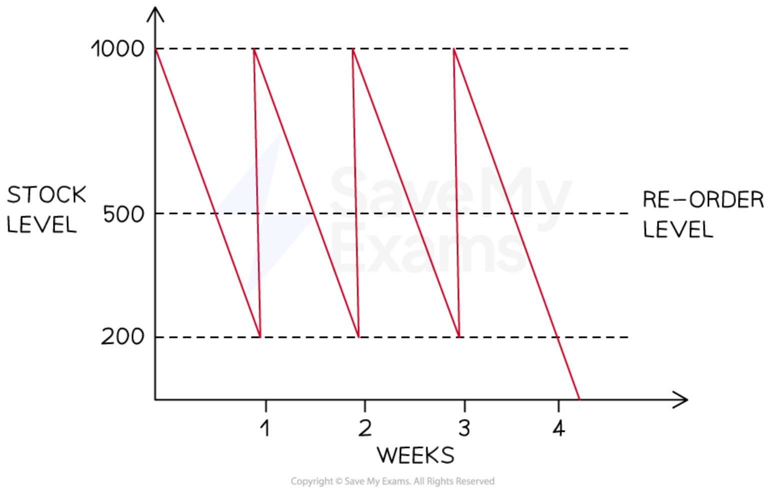
Stock control diagrams show maximum and minimum stock levels, reorder level, lead time and buffer stock level

Diagram analysis

- The **maximum stock level** is the maximum amount of stock a business is able to hold in normal circumstances (1,600)
- The **reorder level** is the level at which a business places a new order with its supplier (800)
- The **minimum stock level** is also known as the **buffer stock** level and is the lowest level to which a business is willing to allow stock levels to fall (400)
- The **lead time** is the length of time from the point of stock being ordered from the supplier to it being delivered (one week)
- The stock level line shows how stock levels change over the given time period
  - As stock is used up, a downward slope is plotted
  - When an order is delivered by a supplier, the stock level line shoots upwards

Worked Example

The diagram below shows stock movements of kitchen shelving units sold by TamFix Ltd.



Graph showing stock level, reorder level and number of weeks for TamFix Ltd

Identify the following points:

1. The minimum stock level
2. The reorder level
3. The reorder quantity
4. The lead time for kitchen shelving units (4)

Step 1: Identify the minimum stock level

The minimum stock level is identified by the bottom-most dotted line. In this case, it shows that the minimum stock level is 200 units. **[1]**

Step 2: Identify the reorder level

The reorder level is clearly identified on the diagram. In this case, it shows that the reorder level is 500 units. **[1]**

Step 3: Identify the reorder quantity

The reorder quantity is the difference between the maximum stock level (shown by the topmost dotted line) and the minimum stock level.

1,000 units (maximum stock) – 200 units (minimum stock) = 800 units

The reorder quantity is therefore 800 units. **[1]**

Step 4: Identify the lead time for kitchen shelving units

The lead time is the difference in time between an order for stock being placed and its delivery. In this case, assuming a five-day working week, the lead time for shelving units is two days. **[1]**

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Buffer stock

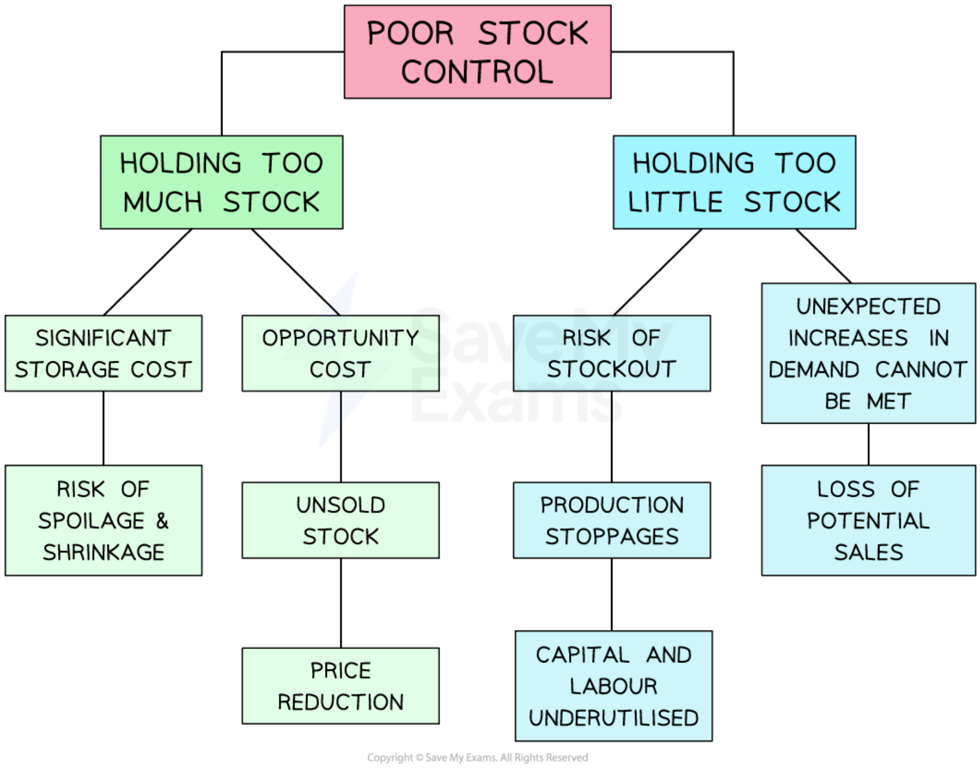
- Buffer stock is a quantity of goods/raw materials kept in case of **stock shortages**
  - This can provide a **competitive edge** over rivals unable to meet demand
  - This approach is commonly called **"just in case"** stock control
- The decision to keep buffer stocks is one that businesses have to weigh up very carefully
  - The decision will be influenced by the nature of the business and the product/service it provides

Advantages and disadvantages of holding buffer stock

Advantages	Disadvantages
- **Stability in supply** Buffer stocks ensure a stable supply of goods, so the business is able to respond to unexpected customer demand - **Price stabilisation** Buffer stocks can help prevent extreme price fluctuations, as they help the market to **avoid shortages**, which would otherwise result in rapid price increases - **Raw materials security** Businesses that are **dependent on particular raw materials** avoid disruption to their supply - **Competitive advantage** By having a reliable supply of goods, businesses can **gain a reputation** for always being able to meet the needs of their customers	- **Cost** Holding buffer stocks can be expensive, as it requires storage facilities and **inventory management systems** - **Risk of obsolescence** Buffer stocks can become obsolete if the demand for a particular product or input declines - **Opportunity cost** Holding buffer stocks **ties up capital** that could be invested in other areas of the business

Implications of poor stock control

Too much or too little stock: implications



Implications of poor stock control include increased storage costs, risk of stockout and inability to meet demand

- Problems may arise from holding **too much stock**
  - **Storage costs** (e.g. warehouse rental, security costs) will be higher than necessary
  - The risk of spoilage and stock shrinkage will be increased, leading to increased costs
- Similarly, holding **too little stock** is risky
  - A business may **run out of stock**, resulting in production stoppages and higher unit costs related to underused capacity
  - A sudden increase in **demand may not be capable of being met**, which leads to a loss of potential sales revenue

Just-in-time stock management

- **Just-in-time stock management** is a process in which raw materials are **not stored on-site**
  - Stock is ordered as required and delivered by suppliers "just in time" for production
- Careful coordination is needed to ensure that raw materials and components are delivered by suppliers **at the moment that they are to be used**

Advantages and disadvantages of just-in-time stock management

Advantages	Disadvantages
▪ **Stockholding costs**, including storage costs, are minimised ▪ Close working relationships are developed with a **small number of trusted suppliers** ▪ **Cash flow is improved**, as money is not tied up in stock ▪ **Unused storage space** is available for productive use ▪ **Teamwork is encouraged**, so employee motivation is likely to be improved	▪ Bulk-buying **economies of scale** are not generally possible ▪ The ability to respond to **unexpected increases in demand** is reduced ▪ **Administrative costs** related to frequent ordering are increased ▪ **Unreliable suppliers** (e.g. late or poor-quality deliveries) can quickly halt production ▪ Significant changes to **organisational structure** and production controls are required

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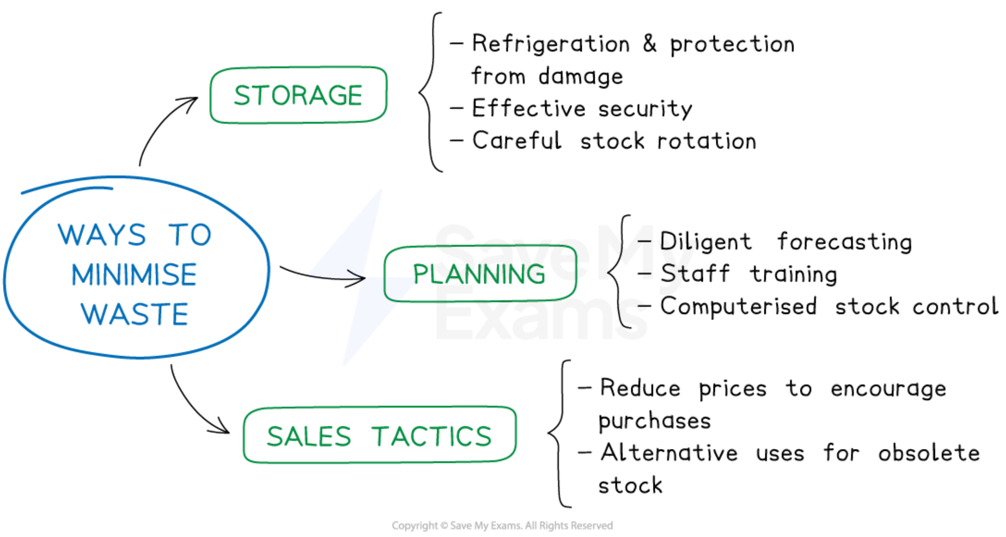
Waste minimisation

- Waste in a business can occur for a number of reasons
  - Stock becomes **obsolete** unless used by a particular date
  - **Perishable** stock (food and medicines) that is not used before it deteriorates will need to be **thrown away**
  - Stock may be **damaged** as a result of poor storage conditions and may **not be suitable for use** in the production process
- Allowing waste to go unchecked will **increase the unit costs of production** and reduce both efficiency and productivity
- Businesses may take a range of steps to **minimise waste**

Ways to minimise waste



Some ways to minimise waste include better storage and planning, as well as changes to sales tactics

- The minimisation of waste will depend on the nature of the product
  - For **perishable items**, refrigeration and careful stock rotation can reduce waste
- Staff training and **computer inventory management systems** may also reduce waste, as fewer errors are likely to be made
- Effective sales forecasting can help reduce the amount of wasted stock

Competitive advantage from lean production

- Lean production involves the **minimisation of the resources** used in production
  - **Less time** is required, as the production process is organised in the most efficient way
  - **Fewer materials** are used, as there is a focus on waste reduction
  - **Less labour** is used, as lean production is typically capital-intensive
  - The **space required for production is reduced** as a result of just-in-time stock management
  - A **small number of trusted suppliers** work closely with the business
- The use of lean production is likely to lead to a competitive advantage
  - Lower **unit costs** are achieved due to minimal wastage, so **prices may be lower** than those offered by competitors
  - **Better quality of output** is likely, as a result of supplier reliability and carefully managed production processes

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Quality management methods

- Quality considers the characteristics and **features** of a product that **satisfy the needs of customers**
- Businesses need to **maintain a level of quality** that continues to attract and retain customers
- There are several approaches to **managing quality** in the production process

Methods of quality management

Method	Explanation

Quality control	▪ Inspecting the quality of output **at the end** of the production process
	▪ Benefits
	  ▪ Quality **specialists** are employed to check standards
	  ▪ An **inexpensive and simple** way to check that the output is fit for purpose
	▪ Drawbacks
	  ▪ The rejection of finished goods is a significant **waste of resources**
	  ▪ There is little focus on the **cause of defects**

Quality assurance	▪ Inspecting the quality of production **throughout** the production process
	▪ Benefits
	  ▪ Quality issues are **identified early**, so **products may be reworked** rather than rejected
	  ▪ The **cause of defects** is the focus, so **future quality issues** may be prevented
	▪ Drawbacks
	  ▪ Staff **training** and a **skilled workforce** are required, so labour costs may be increased
	  ▪ Reworking may **lengthen the production process**

Quality circles	▪ **Groups of workers** meet regularly to **solve quality problems** identified in the production process

	- Benefits
	  - Workers may be **motivated**, as they are **involved in decision-making**
	  - **Relevant and focused solutions** are likely, as workers are familiar with processes
	- Drawbacks
	  - Management needs to have **trust in workers'** views and solutions
	  - **Meetings** must be organised regularly

Total quality management	- Organisation of the business with **quality at its core** and with **every worker responsible for quality**
	- Benefits
	  - Quality in all aspects of the business improves **efficiency**
	  - A **culture of constant improvement** exists within the business
	- Drawbacks
	  - All **workers must be committed** and receive **significant, continuous training**
	  - Careful **monitoring** and **control** are required

Continuous improvement (kaizen)

- Kaizen involves a business taking **continuous steps to improve productivity** through the **elimination of all types of waste** in the production process
  - **Changes are small and ongoing** rather than significant one-offs, and they are **constantly reviewed** to ensure that the desired positive impact on productivity is achieved

Comparison of kaizen and one-off improvements

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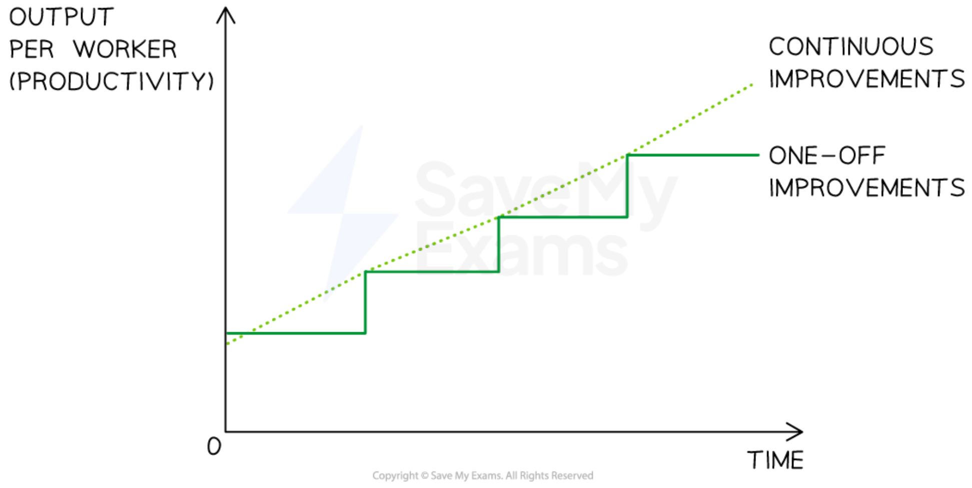

Continuous improvement is likely to lead to higher output per worker over time

- Elements of kaizen commonly include:
  - **Total quality management**
  - Just-in-time stock management
  - Teamwork and quality circles
  - **Zero defects** in manufacturing
  - High levels of automation
  - High levels of cooperation between **workers and management**
- Kaizen requires a **long-term management commitment to change**

Competitive advantage from quality management

- The quality of a business's products can provide a competitive advantage
  - **Unit costs are likely to be low** if a business takes a **preventative approach** through the use of quality assurance or total quality management
    - Low costs may allow a business to **reduce its selling price** to better compete with or undercut its rivals
  - Increased **finance may be available to fund marketing activity** to improve brand recognition and attract new customers
  - High levels of quality can be **used in promotional activity** and provide a **unique selling point** for businesses in competitive markets
  - Successfully developing a unique selling point for quality can **ease expansion into new markets** as a result of the positive reputation it creates

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